

Appendix: Matrix Calculus for the Normal Equations

INFO 521 · Module 1 · Lecture B — supplement

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Adapted with permission from lecture materials by Clayton Morrison, University of Arizona.

Where we are

Lecture B stated the gradient of the least-squares loss directly:

$$\nabla_{\mathbf{w}} \mathcal{L} = -\frac{2}{N} \mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{w}).$$

This appendix fills in the two vector-calculus identities behind it and turns the crank to $\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$.

Two identities

For a constant vector $\mathbf{a}$ and constant matrix $\mathbf{A}$:

$$\frac{\partial (\mathbf{a}^\top \mathbf{w})}{\partial \mathbf{w}} = \mathbf{a}$$

$$\frac{\partial (\mathbf{w}^\top \mathbf{A} \mathbf{w})}{\partial \mathbf{w}} = (\mathbf{A} + \mathbf{A}^\top) \mathbf{w} = 2\mathbf{A} \mathbf{w} \quad (\mathbf{A} \text{ symmetric}).$$

Expand the loss

Write the loss as a quadratic in $\mathbf{w}$:

$$\mathcal{L}(\mathbf{w}) = \frac{1}{N} (\mathbf{y} - \mathbf{X}\mathbf{w})^\top (\mathbf{y} - \mathbf{X}\mathbf{w})$$

Multiply out — the two cross terms are equal scalars ($\mathbf{y}^\top \mathbf{X}\mathbf{w} = \mathbf{w}^\top \mathbf{X}^\top \mathbf{y}$):

$$\mathcal{L}(\mathbf{w}) = \frac{1}{N} (\mathbf{y}^\top \mathbf{y} - 2 \mathbf{w}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{w}^\top \mathbf{X}^\top \mathbf{X} \mathbf{w})$$

Differentiate & solve

Apply the two identities term by term ($\mathbf{a} = \mathbf{X}^\top \mathbf{y}$, $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ symmetric):

$$\nabla_{\mathbf{w}} \mathcal{L} = \frac{1}{N} (-2 \mathbf{X}^\top \mathbf{y} + 2 \mathbf{X}^\top \mathbf{X} \mathbf{w}) = -\frac{2}{N} \mathbf{X}^\top (\mathbf{y} - \mathbf{X} \mathbf{w})$$

Set it to zero — the $-2/N$ drops out:

$$\mathbf{X}^\top \mathbf{X} \hat{\mathbf{w}} = \mathbf{X}^\top \mathbf{y} \quad \implies \quad \boxed{\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}}$$